

SYSC 5108 Exam Study

Assignment 2, Question 7

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Question Summary

Question 7 uses histogram features for small 8×8 handwritten images. Each image is represented by counts of gray levels $0, \dots, 7$, denoted GL-0, $\dots$, GL-7. A logistic regression model classifies digits as either 0 or 1. Some predictions, errors, squared errors, and least-squared-error weight-update terms are missing. Then the question asks for updated weights using LSE, predictions for digits 7 and 8, and a one-sample MLE update.

Course and Textbook Cross-References

- **Chapter 3 slides, Logistic Regression Model:** for $K = 2$, logistic regression uses a linear score $z = \mathbf{w}^T \mathbf{x}$ and probability $p = \sigma(z)$.

- **Chapter 3 slides, Logistic Sigmoid Function:**

$$\sigma(z) = \frac{1}{1 + \exp(-z)}.$$

- **Chapter 3 slides, Alternative Notation:** the MLE or cross-entropy gradient has the clean form

$$-\nabla_w J_{\text{MLE}} = (y - \sigma(z))x.$$

- **Question 6 result:** the LSE-on-probabilities update direction is

$$-\nabla_w J_{\text{LSE}} = (y - \sigma(z))\sigma(z)(1 - \sigma(z))x.$$

- **Goodfellow, Bengio, and Courville, *Deep Learning*:** sigmoid outputs model Bernoulli probabilities; cross-entropy/MLE is generally preferred over MSE/LSE for sigmoid outputs because LSE adds an extra sigmoid derivative factor that can saturate.

Given Model

Use an intercept feature $x[0] = 1$. For one image,

$$z = w[0] + \sum_{j=1}^8 w[j]x[j], \quad \sigma(z) = \frac{1}{1 + \exp(-z)}.$$

The original weights are:

$w[0]$	$w[1]$	$w[2]$	$w[3]$	$w[4]$	$w[5]$	$w[6]$	$w[7]$	$w[8]$
0.309	0.100	-0.152	-0.163	0.191	-0.631	-0.716	-0.478	-0.171

Training Histograms

ID	GL-0	GL-1	GL-2	GL-3	GL-4	GL-5	GL-6	GL-7	Digit
0	31	3	6	2	7	5	6	4	0
1	37	3	1	4	1	3	2	13	1
2	31	3	4	1	8	7	3	7	0
3	38	2	3	0	1	1	5	14	1
4	31	5	3	2	5	2	5	11	0
5	32	6	3	2	1	1	5	14	1
6	31	3	5	2	3	6	2	12	0
7	31	4	3	4	1	5	5	11	0
8	38	4	2	2	2	4	4	8	1
9	38	3	2	3	4	4	1	9	1

Part (a): Missing Model Predictions

For each row, compute

$$z_i = w^T x_i, \quad \sigma(z_i) = \frac{1}{1 + \exp(-z_i)}.$$

The missing predictions are:

ID	z_i	$\sigma(z_i)$	Predicted class
1	-1.8040	0.1414	0
4	-6.3160	0.0018	0
6	-6.6770	0.0013	0

Part (b): Missing Error and Squared Error Values

The table uses

$$\text{error}_i = y_i - \sigma(z_i), \quad \text{s.e.}_i = (y_i - \sigma(z_i))^2.$$

ID	Missing value	Value
0	s.e.	0.0000
5	error	0.9744
7	error	-0.0045
8	s.e.	0.9587

Check: the 0.9744 error belongs to ID 5, because

$$1 - 0.0256 = 0.9744.$$

Part (c): Missing $w[j]$ Error Values

The table defines the LSE update direction as

$$w[j] \text{ error} = -\frac{\partial L}{\partial w[j]} = (y_i - \sigma(z_i))\sigma(z_i)(1 - \sigma(z_i))x_i[j].$$

The missing $w[j]$ error values are:

ID	Weight index	Missing value
3	$w[0]$	0.0503
7	$w[2]$	-0.0001
2	$w[8]$	0.0000

Part (d): New Weights Using LSE

Use learning rate $\eta = 0.01$ and a batch update over all ten training samples:

$$w_{\text{new}}[j] = w_{\text{old}}[j] + \eta \sum_{i=0}^9 (y_i - \sigma(z_i)) \sigma(z_i) (1 - \sigma(z_i)) x_i[j].$$

The summed update directions and new weights are:

Weight	$\sum_i w[j]$	error _i	New weight
$w[0]$	0.2263		0.3113
$w[1]$	8.3484		0.1835
$w[2]$	0.7212		-0.1448
$w[3]$	0.4229		-0.1588
$w[4]$	0.5878		0.1969
$w[5]$	0.3287		-0.6277
$w[6]$	0.5771		-0.7102
$w[7]$	0.6887		-0.4711
$w[8]$	2.8061		-0.1429

Part (e): Predictions for Digits 7 and 8 Using LSE Weights

The new query histograms are:

Digit	GL-0	GL-1	GL-2	GL-3	GL-4	GL-5	GL-6	GL-7
7	35	1	5	4	5	2	4	8
8	30	6	2	0	5	4	4	13

Using the LSE-updated weights from part (d):

True digit	z	$\sigma(z)$	Predicted class
7	-1.0049	0.2680	0
8	-5.0926	0.0061	0

Comment: both out-of-distribution examples are predicted as class 0. This is not very meaningful, because the model was trained only to distinguish digits 0 and 1. Digits 7 and 8 are outside the training classes, so the outputs should only be interpreted as “more like 0 than 1” according to this particular histogram-based model.

Part (f): MLE Update Using Only Sample ID 0

For MLE or cross-entropy, use the update direction

$$-\nabla_w J_{\text{MLE},i} = (y_i - \sigma(z_i)) x_i.$$

The question asks to use only sample ID 0. For ID 0,

$$y_0 = 0, \quad \sigma(z_0) = 0.00010184,$$

so

$$y_0 - \sigma(z_0) = -0.00010184.$$

With $\eta = 0.01$,

$$w_{\text{new}} = w_{\text{old}} + 0.01(y_0 - \sigma(z_0))x_0.$$

Weight	MLE one-sample updated value
$w[0]$	0.3090
$w[1]$	0.1000
$w[2]$	-0.1520
$w[3]$	-0.1630
$w[4]$	0.1910
$w[5]$	-0.6310
$w[6]$	-0.7160
$w[7]$	-0.4780
$w[8]$	-0.1710

The weights appear unchanged to four decimal places because ID 0 is already classified almost perfectly as 0, so $y_0 - \sigma(z_0)$ is extremely small.

Repeating part (e) with these one-sample MLE-updated weights:

True digit	z	$\sigma(z)$	Predicted class
7	-4.2623	0.0139	0
8	-8.0841	0.0003	0

Final Answer

- Missing predictions:

$$\sigma(z_1) = 0.1414, \quad \sigma(z_4) = 0.0018, \quad \sigma(z_6) = 0.0013.$$

- Missing error/squared-error values:

$$\text{s.e.}_0 = 0.0000, \quad \text{error}_5 = 0.9744, \quad \text{error}_7 = -0.0045, \quad \text{s.e.}_8 = 0.9587.$$

- Missing $w[j]$ errors:

$$w[0] \text{ error for ID 3} = 0.0503, \quad w[2] \text{ error for ID 7} = -0.0001, \quad w[8] \text{ error for ID 2} = 0.0000.$$

- LSE-updated weights:

$$(0.3113, 0.1835, -0.1448, -0.1588, 0.1969, -0.6277, -0.7102, -0.4711, -0.1429).$$

- Using the LSE-updated weights, digits 7 and 8 both predict class 0:

$$\sigma(z_7) = 0.2680, \quad \sigma(z_8) = 0.0061.$$

- Using the MLE one-sample ID 0 update, the weights are effectively unchanged at four decimal places, and digits 7 and 8 again predict class 0:

$$\sigma(z_7) = 0.0139, \quad \sigma(z_8) = 0.0003.$$